DO NOW — RECALL

Debugging and error correction

Name: _____ Class: _____ Date: _____

1. Write −1 in 8-bit two's complement.

2. What is the last index in a four-item array?

3. What precondition does binary search need?

4. What happens if code reads a[3] from a three-element C# array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

11111111.

2. Model answer

3.

3. Model answer

Sorted input.

4. Model answer

An IndexOutOfRangeException.

Vocabulary — know these terms

off-by-one error — Loop repeats once too many or few times.

exception — Runtime failure or signal raised by an invalid operation.

trace — Record changing execution state.

precondition — Input requirement an algorithm assumes.

termination — An algorithm eventually ends.

NEW LEARNING

Read, annotate and explain

Check valid indexes

A three-element array has valid indexes 0, 1 and 2.

Follow the condition

When i is 3, i<=a.Length remains true and attempts to read a[3], causing an exception.

Fix and retest

Use i<a.Length, which stops before i becomes an invalid index.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Correct a loop summing [2,4] that uses <= length.

Predict

The array's valid indexes are 0 and 1, not 2.

Explain

Use i < values.Length and the total is 6.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Fix the loop condition i<=a.Length when visiting every array item.

Question 2

When a[mid]>target, what should binary search do to high?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Choose normal, boundary and erroneous indexes when a method accepts only indexes 0–4.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Using low=mid+1 after a[mid]<target can prevent non-shrinking binary-search intervals.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.